

Learning to Decompose and Optimize for Large-Scale Overlapping Problems

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Abstract—Large-scale overlapping problems (LSOPs) pose significant challenges in optimization due to the intricate interactions among the subcomponents. Traditional decomposition-based cooperative co-evolution (CC) algorithms generally rely on predefined rules or randomized grouping strategies, making it difficult to achieve reasonable and effective decomposition tailored to the specific characteristics of LSOPs. To address this, we propose a novel reinforcement learning (RL)-based framework, called RLDO, which utilizes a policy network to learn decomposition knowledge for efficient optimization. During each learning iteration, the policy network processes the features of the LSOP instance and generates the subcomponent interaction probability matrix. By sampling from this matrix, RLDO produces multiple decomposition results, which are subsequently integrated into the CC framework for optimization. The fitness values derived from the optimization process are normalized to serve as reward signals. Using the proximal policy optimization (PPO) algorithm, RLDO updates the policy network based on this feedback, enabling the network to learn more effective decomposition strategies. Finally, the well-trained policy network is capable of effectively decomposing and optimizing LSOPs with various complex topologies. Experimental results demonstrate that the proposed RLDO framework, with its interactive learning of decomposition mechanisms guided by optimization feedback, achieves significantly superior optimization performance compared to state-of-the-art LSOP algorithms.

Index Terms—Large-scale overlapping problems, complex topology structure, cooperative co-evolution, reinforcement learning, proximal policy optimization

I. INTRODUCTION

LARGE-SCALE global optimization (LSGO) problems refer to real-valued optimization tasks involving thousands of decision variables [1, 2]. The excessive number of variables leads to the so-called “curse of dimensionality”, which significantly limits the scalability and efficiency of

existing optimization algorithms [3]. Large-scale overlapping problems (LSOPs), a specific class of problems in the field of LSGO, consist of several interdependent subcomponents [2, 4]. Such problems emerge across various fields of scientific research and practical engineering, such as urban water networks optimization [5], multidisciplinary design optimization [6], supply chain management [7].

The LSOPs are characterized by four key features: topology structure, shared variable character, overlapping degree, and subcomponent weight [8, 9]. The topology structure describes the interaction relationships between subcomponents, which can be classified into line topology, ring topology, and complex topology [8]. Shared variables are classified into two types according to whether their optimal values are identical across adjacent subproblems: conforming overlapping variables and conflicting overlapping variables [2]. Overlapping degree refers to the number of shared variables between adjacent subcomponents, reflecting the intensity of interaction between them [10]. The weights of the subcomponents reflect the relative importance of each subcomponent in the overall problem [11].

In addition to the ubiquitous “curse of dimensionality” common to LSGO [3, 12], LSOPs present unique challenges arising from the complex interaction relationships among overlapping subcomponents. The presence of conflicting overlapping variables results in competition during the optimization of adjacent subcomponents [9]. Moreover, the varying overlapping degrees between subcomponents, along with the potential differences in their respective weights, further complicate the optimization process of LSOPs [10, 11].

Two primary types of approaches have been developed to address these challenges: non-decomposition-based methods and decomposition-based methods [13, 14]. Non-decomposition-based methods generally enhance the exploratory and exploitative capabilities of algorithms by refining genetic operators or combining multiple algorithms. Representative examples of non-decomposition-based methods include the CSO algorithm [15], which improves search operators; the MOS algorithm [16], which combines multiple algorithms; and the SaNSDE algorithm, which employs neighborhood search and adaptive parameters [17].

Decomposition-based methods, inspired by the “divide-and-conquer” principle, tackle high-dimensional problems through a cooperative co-evolution (CC) framework [18–20]. However, the overlapping characteristics of LSOPs introduce difficulties for mainstream decomposition-based algorithms in LSGO [21–23]. Most methods, such as ERDG [24], GSG [25] and

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IRRG [26], group all interacting variables into a single variable group, thereby hindering effective decomposition. These approaches fail to capture the underlying structure and modular characteristics of LSOPs, resulting in suboptimal optimization performance. Extensive experiments have demonstrated that exploiting the inherent modular structure of optimization problems to perform rational decomposition constitutes an effective strategy for enhancing optimization performance [10, 11, 27]. To achieve decomposition for LSOPs, researchers have proposed two primary types of decomposition-based optimization methods: linkage breaking-based optimization methods [10, 28, 29] and interaction structure matrix-based optimization methods [11, 30, 31].

For example, to overcome the limitations of finite-difference-based algorithms [24, 32], RDG3 [10] breaks the linkages between subcomponents by imposing a subcomponent size threshold and randomly assigns shared variables to adjacent subcomponents, thus decomposing the LSOPs. However, this approach cannot guarantee the rationality or effectiveness of the decomposition. Similarly, ORDG [28] attempts to identify and iteratively optimize each subcomponent, but suffers from randomness in subcomponent identification, leading to low accuracy and poor stability, and does not explore more efficient subcomponent partitioning during optimization.

To detect the underlying variable interaction structure in LSOPs, several methods such as GDG [33] and DG2 [34] construct interaction matrices by identifying pairwise variable dependencies. However, their computational complexity of $\mathcal{O}(N^2)$, where N denotes the number of decision variables, can result in substantial computational cost, especially for high-dimensional problems. To address this limitation, the OEDG method [8] introduces a more efficient two-stage approach with a reduced complexity of $\mathcal{O}(N \log N)$, making it more scalable for large-scale applications. Additionally, methods like CBCCO [11] provide effective weight estimation by evaluating the contribution of each subcomponent to the fitness improvement in early-stage optimization, and use this insight to refine the allocation of shared variables. Yet, CBCCO does not provide a comprehensive global decomposition strategy.

It is worth noting that, except for OEDG, most of these methods were designed for LSOPs with line topologies [11, 28]. The concept of large-scale complex topological overlapping problems were introduced in [8] and presents additional challenges. In complex topologies, one subcomponent may interact with multiple others, resulting in larger independent variable groups when using linkage-breaking methods, and rendering interaction structure matrix-based approaches less effective at allocating shared variables due to inherent uncertainties.

In summary, while structural identification methods for LSOPs, including variable interaction detection and subcomponent weight estimation, are relatively well developed [11, 28, 34], there exists a significant gap in the research on optimization methods for LSOPs with complex topologies. Existing approaches are predominantly designed for line topology problems and rely on sequential or random decomposition, which fail to consider the unique structural characteristics

of LSOPs with complex topologies. Experimental results in [8] suggest that even random decomposition-based methods can outperform non-decomposition-based methods in these settings, indicating that an appropriate degree of decomposition can enhance optimization performance. Therefore, it is urgent to explore more reasonable and efficient decomposition-optimization methods tailored for such problems with complex topologies by building them upon existing research on accurate structural identification for LSOPs.

Recently, learning-based approaches, particularly reinforcement learning (RL), have received increasing attention for assisting evolutionary optimization [35, 36]. Leveraging its ability to develop strong discriminative guidance and decision-making capabilities, RL has been applied to improve various components of evolutionary algorithms (EAs), such as variation operator adaptation [37, 38], dynamic operator selection [39, 40], and dynamic hyperparameter configuration [41, 42]. Furthermore, RL has been explored for automated algorithm design, including algorithm selection [43, 44], algorithm configuration [45, 46], and algorithm generation [47]. These studies demonstrate that RL can enhance the performance of EAs by learning intelligent strategies through trial-and-error interactions with the environment, suggesting its potential to learn effective decomposition strategies for complex topological LSOPs.

This work proposes a reinforcement learning-based decomposition and optimization framework, termed RLDO. By leveraging the interactive learning capabilities of reinforcement learning (RL), we develop a decomposition policy network specifically for complex topological overlapping problems. The policy network iteratively learns decomposition strategies, guided by optimization feedback from various decomposition results. In the training phase, the policy network generates a subcomponent interaction probability matrix based on the features of the input LSOP instance. Sampling from this matrix produces multiple decomposition outcomes, which are then integrated into a cooperative co-evolution optimization framework. The fitness values obtained from the optimization process are normalized into reward signals that update the policy network. Through continuous learning from decomposition experiences, the well-trained policy network efficiently decomposes and optimizes LSOPs with diverse complex topological structures. The main contributions of this work are as follows:

- To the best of our knowledge, this work makes the first attempt to apply reinforcement learning to bridge the research gap in the optimization of large-scale complex topological overlapping problems, enabling effective decomposition and optimization within the cooperative co-evolution framework.
- Unlike traditional methods that rely on predefined or random decomposition, RLDO dynamically learns optimal decomposition policies through exploration and feedback. The policy network generates a subcomponent interaction probability matrix at each training iteration, and optimization feedback is transformed into reward signals for network updates.
- Extensive experiments on two large-scale complex topological overlapping problem benchmarks demonstrate that

RLDO achieves reasonable and effective decomposition and optimization, outperforming state-of-the-art methods.

The remainder of this article is organized as follows. Section II presents the relevant background on LSOP and reinforcement learning. Section III introduces the proposed RLDO framework in detail. The experiments are presented in Section IV. Section V concludes this article and outlines future research directions.

II. BACKGROUND

In this section, we first introduce the preliminaries related to LSOPs, including fundamental concepts, underlying structure identification methods, and optimization methods. Subsequently, we provide a brief introduction to reinforcement learning.

A. Large-scale Overlapping Problems

1) *Definition of LSOPs*: LSOPs are a class of large-scale global optimization problems with special variable interaction structures [2, 4]. Without loss of generality, an LSOP can be mathematically formulated as follows:

$$\begin{aligned} \min_{\mathbf{x}} f(\mathbf{x}) &= f(x_1, x_2, \dots, x_n) \\ \text{s.t. } \mathbf{x} &\in \Omega \subseteq \mathbb{R}^n. \end{aligned} \quad (1)$$

Here, the variables $(x_1, x_2, \dots, x_n)$ are partitioned into multiple overlapping subcomponents $(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m)$, with all variables within each subcomponent interacting directly with one another. Shared variables—those simultaneously existing in multiple subcomponents—result in direct or indirect interactions among any two subcomponents in an LSOP.

According to the definition of separability, overlapping problems exhibit a non-separable property characterized by a modular interaction structure with overlapping subcomponents. We present a diagram, illustrated in Fig. 1, to provide a simple example of an overlapping problem. This problem consists of two subcomponents: $\mathbf{G}_1 = \{x_1, x_2\}$ and $\mathbf{G}_2 = \{x_2, x_3\}$, containing shared variables x_2 and non-shared variables x_1 and x_3 . The variables x_1 and x_2 in $\mathbf{G}_1$, as well as x_2 and x_3 in $\mathbf{G}_2$, interact directly. Additionally, x_1 and x_3 interact indirectly through x_2 , which can be denoted by $x_1 \leftrightarrow x_2 \leftrightarrow x_3$.

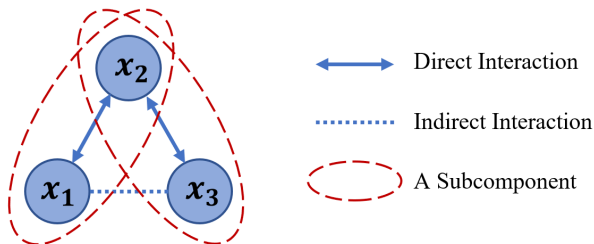


Fig. 1: Illustration of an overlapping problem $f(\mathbf{x}) = (x_1 - x_2)^2 + (x_2 - x_3)^2$.

2) *Problem Properties of LSOPs*: A defining characteristic of LSOPs is the complex interactions among variables. LSOPs exhibit four key properties: topology structure, overlapping degree, subcomponent weight, and shared variable character.

The topology structure reflects the interaction relationships among subcomponents, specifically indicating which subcomponents share common variables. These interactions can be represented by a subcomponent interaction structure matrix $\mathbf{M}$, where $M_{ij} = 1$ denotes that subcomponent i and subcomponent j interact.

The overlapping degree quantifies the extent of interaction between neighboring subcomponents by indicating the number of shared variables between them. It is defined as:

$$OD = \frac{k}{n}, \quad (2)$$

where n represents the total number of variables in the problem, and k denotes the number of shared variables. The overlapping degree can be further incorporated into the subcomponent interaction structure matrix $\mathbf{M}$, such that $M_{ij} = m$ indicates that subcomponent i and subcomponent j share m variables.

In many LSOPs, different subcomponents contribute unequally to the overall fitness improvement when optimized under a CC framework. This imbalanced importance is often modeled by assigning different weights to each subcomponent [1].

Shared variables can be categorized into two types: conforming overlapping variables and conflicting overlapping variables. Conforming overlapping variables have the same optimal value across all adjacent subcomponents. Therefore, optimizing one subcomponent allows the optimized shared variables to positively contribute to other adjacent subcomponents. In contrast, conflicting overlapping variables exhibit different optimal values for each subcomponent. Optimizing one subcomponent might adversely affect other overlapping subcomponents, making conflicting overlapping problems inherently more challenging to solve than conforming overlapping problems.

3) *Structure Identification Methods of LSOPs*: To address the challenges posed by LSOPs, researchers have proposed a series of methods for identifying the underlying structure to facilitate optimization [10, 28, 34].

RDG3 introduces a threshold to prematurely terminate the merging process for subcomponents identification. However, RDG3 is limited to identifying only partial subcomponents and fails to accurately analyze the overall structure of LSOPs. ORDG attempts to iteratively identify each subcomponent using shared variables, thereby achieving the identification of its underlying structure. However, ORDG is limited in handling LSOPs with simple topology structures, and the accuracy of its underlying structure identification heavily depends on the selection of initial detected variable.

To obtain the global structural information of LSOP, DG2 detects the pairwise interactions between all variables and constructs an interaction structure matrix. Despite its effectiveness, DG2 demands excessive computational resources for interaction detection. To address this, OEDG introduces a two-

stage recursive interaction detection method that efficiently and accurately identifies the underlying structure of LSOPs.

4) *Optimization Methods of LSOPs*: A significant research gap persists in developing optimization methods for LSOPs with complex topologies, as most existing work focuses on LSOPs with line topologies [11, 30, 31]. Several approaches have been proposed.

RDG3 randomly assigns shared variables to adjacent sub-components, decomposing the LSOP into multiple independent variable groups. These groups are subsequently incorporated into the CC framework for optimization. Additionally, DCC [48] generates subgroups utilizing optimization information from the randomized grouping optimization process. However, it overlooks the modular characteristics of LSOP, resulting in low optimization efficiency.

ORDG attempts to leverage the accurate information of each subcomponent and sequentially optimizes each subcomponent. During the optimization of each subcomponent, ORDG optimizes all shared and non-shared variables within the subcomponent. To determine the value of the shared variable x_i in the context vector, ORDG compares all individuals in the subpopulations containing x_i , assigning the value based on the best-performing individual.

Based on the interaction structure matrix of LSOPs, CBCCO allocates shared variables to neighboring subcomponents according to their contributions, which are determined by the fitness value improvement of non-shared variables in each subcomponent during the early stages of optimization. After forming independent variable groups, CBCCO employs a reward mechanism to allocate more computational resources to subcomponents with larger contributions during the CC optimization process. Additionally, some methods [30, 31] design novel computational resource allocation schemes to enhance the optimization efficiency of overlapping problems. However, they also follow to the sequential decomposition CC framework.

B. Reinforcement Learning

RL is a machine learning paradigm in which an agent interacts with an environment to learn a policy that maximizes cumulative rewards [49–51]. This interaction is typically formulated as a Markov decision process, where the agent observes a state s_t , takes an action a_t following a policy $\pi(a_t|s_t)$, and receives a scalar reward $r(s_t, a_t)$. The learning objective is to maximize the expected discounted return:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]. \quad (3)$$

RL algorithms can be broadly divided into model-based and model-free approaches [49]. Model-based methods construct approximate environment dynamics to guide policy learning, while model-free methods enable efficient policy learning through direct interaction with the environment, offering greater flexibility and simplicity in implementation [52, 53].

Proximal policy optimization (PPO) [54] is a well-established RL algorithm that has been widely applied across various domains [55–57]. PPO uses importance sampling to

reuse past experience, thereby reducing reliance on interaction data and improving training efficiency. PPO minimizes large policy updates via a clipped surrogate objective:

$$\mathcal{L}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (4)$$

where $r_t(\theta) = \pi_{\theta}(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ is the probability ratio between the updated and previous policies, $\hat{A}_t$ is the advantage estimate, and ϵ is a hyperparameter that controls the clipping range. This approach supports efficient learning with stable policy updates, making it suitable for guiding decomposition strategy learning in our framework.

III. THE PROPOSED RLDO

To address the challenges of existing methods that fail to efficiently and rationally decompose and optimize large-scale complex topological overlapping problems, we propose a reinforcement learning-based decomposition and optimization (RLDO) framework¹. Within the RL framework, RLDO learns the optimal decomposition policy model by leveraging feedback from the optimization results across various decomposition outcomes under the CC framework. The well-trained decomposition policy model is then used to achieve effective and rational decomposition for LSOP instances with diverse complex topologies.

In this section, we first introduce the overall framework of RLDO. Next, we present the details of the decomposition policy network model, the data collection process, and the proximal optimization policy designed within the RLDO framework. Finally, we present the process of applying the well-trained model to perform decomposition and optimization for large-scale complex topological overlapping problems.

A. The Overall Framework of RLDO

The RLDO framework consists of two stage: policy network training and policy network-guided optimization. We present a diagram, shown in Fig. 2, to visually illustrate the complete structure of the RLDO framework.

In the policy network training phase, the decomposition and optimization process of LSOP is integrated into the RL framework, enabling the policy network to learn effective decomposition strategies. Specifically, the states are defined as overlapping problems, characterized by varying topology structures and subcomponent weights. The actions correspond to the different decomposition results. The rewards are defined as the optimization results obtained from the CC framework for each decomposition result, reflecting the quality of the decomposition action. Each LSOP instance, its decomposition result, and its corresponding optimization performance form a state-action-reward tuple, which the RL model learns from.

To effectively learn decomposition strategies for LSOPs with complex topologies, we adopt a Transformer-based policy network due to its superior feature extraction capabilities, which are well-suited for handling overlapping variable interactions.

¹The source code: <https://github.com/tian2156/RLDO>

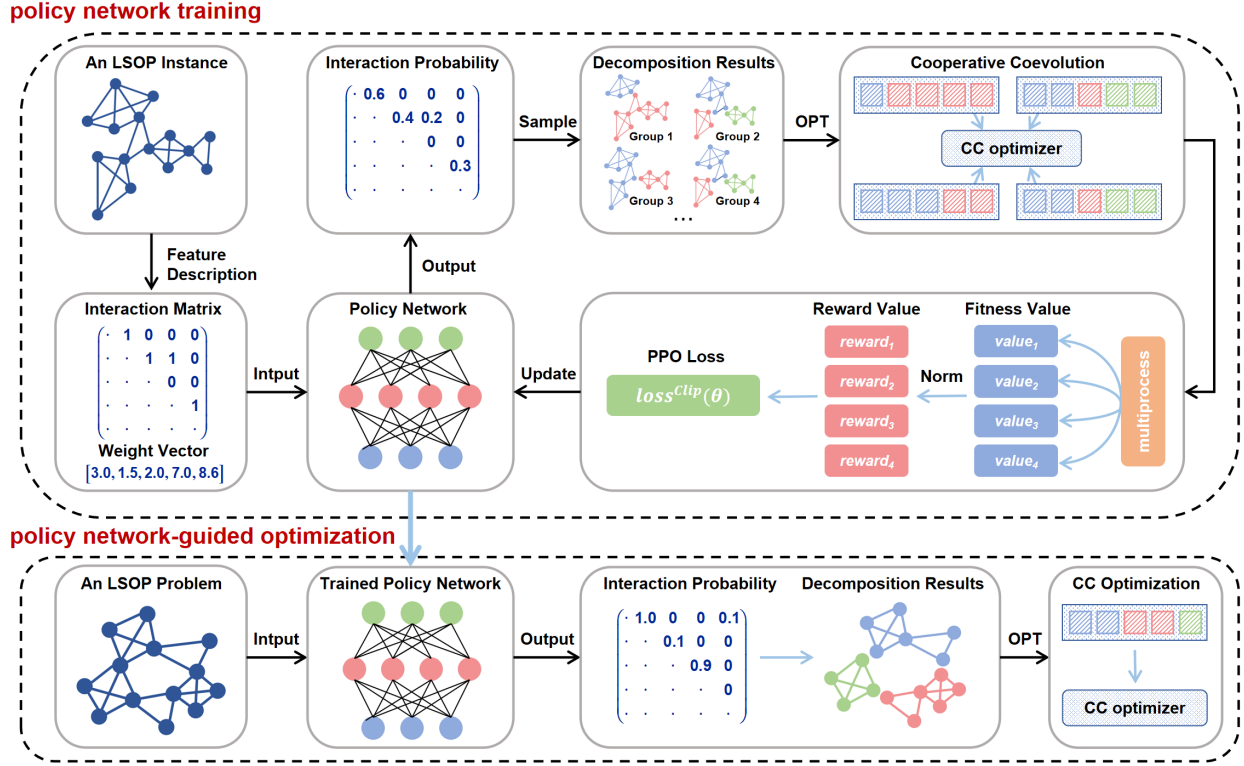


Fig. 2: The overall flowchart of the RLDO framework.

For policy network training, we begin by collecting interaction data, which consists of LSOP instances, decomposition actions, and corresponding optimization results. To generate diverse LSOP instances with complex topologies, we employ the Complex Topology Overlapping Construction (CTOC) method [8]. The features of these LSOP instances are then fed into the policy network, which generates a subcomponent interaction probability matrix for each instance. This matrix is subsequently sampled to produce multiple decomposition results. Once the decomposition results are obtained, they are integrated into the CC framework for optimization. The optimization process yields fitness values, which are then normalized into reward signals to ensure a consistent evaluation metric. These reward signals help guide the learning process of the policy network.

After collecting sufficient interaction data, we update the policy network using the PPO algorithm. PPO enhances the effective use of limited data by performing multiple policy updates and ensures stable network updates through its clipping mechanism. This stability is crucial for preventing excessive fluctuations in policy updates and maintaining consistent learning progress.

In the policy network-guided optimization phase, the well-trained policy network can effectively guide the decomposition and optimization of overlapping problems with various topologies. Given an LSOP instance, the network predicts the subcomponent interaction probability matrix, which determines the optimal decomposition strategy. The decomposition result is then integrated into the CC optimization framework for further optimization, ensuring an efficient and structured

approach to solving LSOPs.

The following subsections will provide a detailed explanation of the policy network model for LSOP decomposition, data collection, PPO algorithm in RLDO framework, and policy network-guided optimization.

B. Policy Network Model for LSOP Decomposition

In the RLDO framework, the policy network is constructed to learn decomposition knowledge during training, enabling efficient LSOP decomposition and enhancing problem-solving performance.

The input to the decomposition policy network includes a subcomponent adjacency matrix, denoted as $M \in \mathbb{R}^{n \times n}$, and a subcomponent weight vector, $V \in \mathbb{R}^n$, where n represents the number of subcomponents in LSOP. Each element M_{ij} in matrix M captures the number of shared variables between subcomponents i and j , while the weight vector V denotes the weight of each subcomponent. The network's output is a subcomponent interaction probability matrix $W \in \mathbb{R}^{n \times n}$, where a non-zero element W_{ij} denotes the probability that subcomponents i and j interact under the optimal decomposition.

To process the input, three types of information are incorporated: the adjacency matrix $M \in \mathbb{R}^{n \times n}$, the subcomponent weight vector $V \in \mathbb{R}^n$, and the positional index vector $N \in \mathbb{Z}^n$. The matrix M is passed through a linear projection layer to produce a topology-based embedding $M' \in \mathbb{R}^{n \times d}$. The vector V is normalized and transformed via a linear layer to obtain a weight embedding $V' \in \mathbb{R}^{n \times d}$. The index vector N , which assigns each subcomponent a unique index from

1 to n , is embedded using a categorical embedding layer to produce $N' \in \mathbb{R}^{n \times d}$. These three embeddings— M' , V' , and N' —are then fused through element-wise addition to form a unified representation $X \in \mathbb{R}^{n \times d}$.

The fused information X is further processed through two layers of Transformer encoder blocks, which capture complex interdependencies and extract higher-order feature representations. The output layer then transforms the extracted features into a raw subcomponent interaction probability matrix, reflecting the strength of relationships between subcomponents. To ensure logical consistency in decomposition, the raw probability matrix is masked based on the input interaction matrix M , preventing the introduction of new interactions between previously unconnected subcomponents. A Softmax operation is finally applied to normalize the raw probabilities into meaningful subcomponent interaction probabilities.

Algorithm 1 Data Collection

Input: B – the problem batch size
 S – the number of sampled groups
 π_θ – current policy network
Output: D – memory buffer in PPO

- 1: $P_i \leftarrow \text{Creat LSOP by CTOC, } i = 1, 2, \dots, B$
- 2: $W_i \leftarrow \pi_\theta(P_i)$
- 3: $G_i^j \leftarrow \text{Sample from } W_i, j = 1, 2, \dots, S$
- 4: $f_i^j \leftarrow \text{DECC}(P_i, G_i^j)$
- 5: $f_i^j \leftarrow -1 \times f_i^j$
- 6: $R_i^j \leftarrow \text{Normalize the } f_i^j \text{ based on Equation (5)}$
- 7: $D \leftarrow \text{Save interaction data } \{P_i, G_i^j, R_i^j\}$
- 8: **return** D

C. Data Collection

Data collection in RLDO is described in Algorithm 1. In each iteration, we referred to the CTOC method to generate B instances of LSOP (Line 1). The interaction probability is set to 0.2, consistent with the prior configuration in the original CTOC method. To introduce different interaction degrees among subcomponents, the overlapping size between interacting subcomponents is adjusted from a fixed value in the CTOC method to a random integer ranging from 1 to 10. Other problem settings remain consistent with those in [8].

The policy network takes the subcomponent interaction matrix M and weight vector V of each LSOP instance P_i as input and outputs the corresponding interaction probability matrix W_i (Line 2). This subcomponent interaction probability matrix is then repeatedly sampled to generate multiple decomposition results (Line 3).

Next, each grouping result G_i^j is integrated into the CC framework for optimization (Line 4). We employ the fundamental CC algorithm as the optimization framework, with the basic Differential Evolution (DE) algorithm as the optimizer. Since the optimization processes for different LSOP instances across various decomposition results are independent, we implemented a multi-process parallel computing framework, significantly reducing computation time.

The CTOC method randomly generates LSOPs with varying topological structures and subcomponent weights, leading to

substantial differences in optimization outcomes and problem difficulty across instances. As a result, the fitness values obtained from different instances may fall into different intervals, requiring normalization to establish a unified reward criterion. For each LSOP instance, the fitness values $f_i = \{f_i^1, f_i^2, \dots, f_i^S\}$ are normalized to the reward information in the range of $[-1, 1]$ according to the following formula:

$$R_i^j = \frac{2 \times (f_i^j - \text{mean}(f_i))}{\max(f_i) - \min(f_i)} - 1. \quad (5)$$

D. PPO Algorithm in RLDO Framework

We apply the PPO method to optimize the network parameters by leveraging the experience data collected in the buffer. The pseudocode of policy network update based on PPO algorithm can be found in Algorithm 2.

The PPO algorithm is capable of conducting multiple updates utilized a set of data and integrating the clipped objective function to guarantee the stability of policy updates. This method enhances the effective use of limited data and alleviates the difficulty of excessive computational resource consumption of LSOP.

The loss function for PPO in RLDO is defined as follows:

$$\mathcal{L}(\theta) = \mathbb{E}[\min(r(\theta)R, \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)R)] + \beta \mathbb{E}[\mathcal{H}(\pi_\theta)], \quad (6)$$

$$\mathcal{H}(\pi_\theta(\cdot | s)) = - \sum_{\mathbf{G}} \pi_\theta(\mathbf{G} | \mathbf{P}) \log(\pi_\theta(\mathbf{G} | \mathbf{P})), \quad (7)$$

where $r_t(\theta) = \pi_\theta(\mathbf{G} | \mathbf{P}) / \pi_{\theta_{\text{old}}}(\mathbf{G} | \mathbf{P})$ is the probability ratio between new and old policies. To ensure stable policy updates, the clipping term restricts the policy ratio within a small range, with ϵ set to 0.1 in our implementation. Additionally, an entropy term $\mathcal{H}(\pi_\theta(\cdot | \mathbf{P}))$ is introduced to encourage exploration by the policy network.

Algorithm 2 Policy Network Update Based on PPO

Input: $\pi_{\theta_{\text{old}}}$ – the old policy network
 D – the collected data buffer
 n – the number of update epochs
 ϵ – the clipping coefficient
 β – the entropy coefficient

Output: $\pi_{\theta_{\text{new}}}$ – the updated policy network

- 1: **for** $e \leftarrow 1$ to n **do**
- 2: **for** each mini-batch data $\subset D$ **do**
- 3: $r_t(\theta) = \pi_\theta(\mathbf{G} | \mathbf{P}) / \pi_{\theta_{\text{old}}}(\mathbf{G} | \mathbf{P})$
- 4: $J_{\text{PPO}}^{\text{clip}}(\theta) = \min(r(\theta)R, \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)R)$
- 5: $H(\pi_\theta) = - \sum \pi_\theta(\mathbf{G} | \mathbf{P}) \log \pi_\theta(\mathbf{G} | \mathbf{P})$
- 6: $J(\theta) = \mathbb{E}[J_{\text{PPO}}^{\text{clip}}(\theta) + \beta H(\pi_\theta)]$
- 7: $\theta \leftarrow \text{Adam}(\theta, \nabla_\theta J(\theta))$
- 9: **end for**
- 10: **end for**
- 11: **return** $\pi_{\theta_{\text{new}}}$

Proximal Policy Optimization (PPO) is adopted for its stability, ease of implementation, and strong empirical performance across reinforcement learning tasks and learning-assisted evolutionary computation. Additionally, in large-scale

optimization problems where function evaluations are costly, PPO can perform multiple stable policy updates on the same dataset, thereby making more effective use of limited data. Finally, RLDO is an algorithmic framework with modular, replaceable components, which facilitates the exploration of alternative training algorithms that may better suit different optimization tasks. While PPO has been chosen as a practical instance for this work, the choice of training algorithm remains open for further discussion.

E. Policy Network-Guided Optimization

Once trained, the policy network can effectively guide LSOP optimization regardless of the complexity of its topological structure. The optimization process under the RLDO framework comprises two stages: the problem decomposition stage (Lines 1–3) and the cooperative co-evolution (CC) optimization stage (Lines 4–15), as shown in Algorithm 3.

The input to the policy network comprises the subcomponent adjacency matrix and subcomponent weight vector, which describe the structural properties of the LSOP instance. These inputs are not assumed to be known a priori; rather, they can be efficiently extracted using established structural analysis techniques such as OEDG [8] for interaction detection and CBCCO [11] for subcomponent weighting. These tools serve as pre-processing modules that enable RLDO to generalize to unseen problems without requiring manual specification of problem structure.

By leveraging these structural descriptors as inputs, RLDO learns a policy that enables adaptive decomposition tailored to the topological characteristics of each problem. This approach extends prior decomposition strategies by incorporating learned structural awareness into the optimization pipeline.

In the problem decomposition stage, the policy network outputs a subcomponent interaction probability matrix $\mathbf{W}$, where each element $\mathbf{W}_{ij}$ represents the likelihood that subcomponents i and j should be grouped together. Higher values of $\mathbf{W}_{ij}$ indicate stronger linkage, while lower values suggest separation. During training, sampling from $\mathbf{W}$ promotes exploration. During inference, binary decisions are made: subcomponents i and j are grouped if $\mathbf{W}_{ij} > 0.5$, and separated otherwise. This process yields a final set of grouped subcomponents for optimization.

In the CC optimization stage, the decomposition results are used in a classical CC framework, where optimization is performed independently on each group. For generality and simplicity, we adopt the DECC framework with the SaNSDE optimizer [22, 23, 32, 34], and allocate equal computational resources to all subcomponents. After initializing the population, the algorithm iteratively refines each group until the termination condition is met.

IV. EXPERIMENTAL STUDIES

This section presents the experimental evaluation of the RLDO framework. We begin by introducing the large-scale complex topological overlapping benchmark problems. Next, we detail the parameter settings used in RLDO. We then describe the comparison algorithms and the experimental setup.

Algorithm 3 Policy Network-Guided Optimization

Input: $\mathbf{P}$ – the large-scale overlapping problem
 π_θ – the well-trained policy network

Output: $best, best_val$

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1: /***** Problem Grouping Stage *****/
2:  $\mathbf{W} = \pi_\theta(\mathbf{P}(\mathbf{M}, \mathbf{V}))$ 
3:  $groups \leftarrow \overset{\mathbf{W}}{\sim} \{\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_n\}$ 
4: /***** CC Optimization Stage *****/
5:  $pop \leftarrow \text{rand}(popsize, n)$ 
6:  $(best, best\_val) \leftarrow \min(func(pop))$ 
7: while the termination condition is not met do
8:   for  $j \leftarrow 1$  to  $size(groups)$  do
9:      $indices \leftarrow groups[j]$ 
10:     $subpop \leftarrow pop[:, indices]$ 
11:     $subpop \leftarrow \text{SaNSDE}(best, subpop, FE)$ 
12:     $pop[:, indices] \leftarrow subpop$ 
13:     $(best, best\_val) \leftarrow \min(func(pop))$ 
14:   end for
15: end while
16: return  $best, best\_val$ 

```

The optimization performance of RLDO is evaluated through comparative experiments on two categories of benchmark problems: conforming overlapping benchmark problems and conflicting overlapping benchmark problems. Furthermore, we analyze the decomposition mechanisms of RLDO and the comparison algorithms. Finally, we evaluate RLDO and the algorithms under comparison on large-scale overlapping problems of different scales.

A. Benchmark

The CEC 2013 LSGO [2] first introduced large-scale overlapping benchmark problems, including examples of conforming and conflicting overlapping problems with line topology. Building on this, OEDG [8] provided an in-depth analysis of LSOP characteristics and introduced the Complex Topology Overlapping Construction (CTOC) method to generate overlapping problems with diverse topological structures.

In this study, we use the CTOC method to construct large-scale complex topological overlapping problems for evaluating the performance of the compared algorithms. Following the settings of CTOC outlined in the original study [8], we constructed both conflicting and conforming benchmark problems. In our setup, the number of subcomponents was set to 10, the group size to 100, and the overlapping degree to 10. The pseudocode of CTOC is presented in Algorithm 4.

The CTOC method begins by initializing the subcomponent interaction structure matrix $\mathbf{M}$ and the set of subcomponents $\mathbf{G}$ and (Line 1). The matrix $\mathbf{M}$ is used to represent the interaction relationship and overlapping degree between the subcomponents, while $\mathbf{G}$ is constructed based on matrix $\mathbf{M}$ to represent the indices of variables within different subcomponents. For the upper triangular part of $\mathbf{M}$, the interaction between subcomponents is determined with probability p (Lines 2–8). The elements on the matrix $\mathbf{M}_{ij}$ represent the number of overlapping variables between subcomponents i

and j . Then, Depth-First-Search (DFS) is utilized to verify the connectivity of the matrix M and identify the connected node set v and the unconnected node set u (Line 9). If the adjacency matrix is not connected, CTOC randomly select one node from v and one node from u to construct a connection between them (Lines 10-19). Finally, the subcomponents set G is constructed based on the interaction relationships and the number of overlapping variables represented in M (Line 20).

Algorithm 4 CTOC

Input: n (number of subcomponents), s (size of subcomponents), m (overlapping size), p (overlapping probability)
Output: G (set of subcomponents)
1: $M \leftarrow \text{zeros}(n, n)$, $G \leftarrow \{\}$
2: **for** $i \leftarrow 1$ to $n - 1$ **do**
3: **for** $j \leftarrow i + 1$ to n **do**
4: **if** $\text{rand}(0, 1) < p$ **then**
5: $M_{ij} \leftarrow \text{rand}(1, m)$
6: **end if**
7: **end for**
8: **end for**
9: $(v, u) \leftarrow$ Use DFS to check the connectivity of M
10: **while** u is not empty **do**
11: $i \leftarrow$ Select one node from v at random
12: $j \leftarrow$ Select one node from u at random
13: **if** $i < j$ **then**
14: $M_{ij} \leftarrow \text{rand}(1, m)$
15: **else**
16: $M_{ji} \leftarrow \text{rand}(1, m)$
17: **end if**
18: $(v, u) \leftarrow$ Use DFS to check the connectivity of M
19: **end while**
20: $G \leftarrow$ Construct the set of subcomponents based on M
21: **return** G

Additionally, after obtaining the set of subcomponents, we implement experimental settings including index shuffling, coordinate shifts, and fitness landscape rotation, as described in the CEC LSGO [1, 2, 58]. A random sequence $\mathcal{P}$ is applied to permute the dimension indices. The optimal location of each decision variable is determined based on the optimal decision vector $\mathbf{x}^{\text{opt}}$. Subsequently, the orthogonal rotation matrix $\mathbf{R}$ is used to randomly rotate the fitness landscape to generate non-separable variable groups. Finally, the subcomponents are then assigned different weights using the random weight generation method in CEC 2013 LSGO to introduce an imbalance effect [2, 58].

B. Parameter Setting

As an algorithm framework with strong scalability and substitutability, RLDO adopts parameter settings primarily based on default values from previous studies.

The training process of the policy network model in RLDO consists of 200 iterations, where each iteration utilizes the CTOC method to randomly generate eight LSOP instances. For each LSOP instance, seven decomposition results are sampled from the interaction probability matrix.

For optimization, RLDO employs the DECC framework as the base CC framework [17], with the DE algorithm as the optimizer [59]. The parameters of the DE algorithm are set as follows: scaling factor $F = 0.5$, crossover rate $CR = 0.3$, and the population size $NP = 50$.

The policy network in RLDO is based on a fundamental Transformer architecture [60], consisting of 2 stacked encoders with 8 attention heads and a model hidden dimension of 512. The learning rate is set to 1×10^{-4} .

The PPO algorithm utilizes a clipping coefficient of 0.1 and performs ten updates using a set of collected data. The mini-batch size is set to 8, and the entropy coefficient is set to 3 to encourage exploration during learning.

C. Experimental Settings

To evaluate the performance of RLDO, we compare it against state-of-the-art methods for LSOPs, including sequential decomposition, random decomposition, and non-decomposition-based algorithms.

The decomposition-based methods are implemented within the standard cooperative co-evolution (CC) framework, using SaNSDE as the optimizer. RDG3 breaks the linkage between subcomponents by applying a subcomponent size threshold, with two variants tested: RDG3-I, using a threshold of 200 (twice the base subcomponent size) and RDG3-II, using a threshold of 400 (four times the base subcomponent size). Additionally, a random decomposition method is introduced, where each non-zero element in the subcomponent interaction matrix is independently set to zero with a probability of 0.5 for decomposition, referred to as Random.

To evaluate the necessity and impact of proper decomposition, we include three representative non-decomposition optimizers: SaNSDE, CSO, and MMES. Each is integrated into RLDO as the subcomponent optimizer, yielding RLDO-SaNSDE, RLDO-CSO, and RLDO-MMES, respectively. The parameter settings are adopted from those reported in the original literature [15, 17, 61]. This configuration thus allows us to evaluate whether learned, problem-specific decomposition improves the performance of non-decomposition optimizers.

The number of function evaluations (FEs) for each problem is set to 3×10^6 as the termination criterion. The optimization experiments for RLDO and other decomposition-based methods aim to investigate the impact of different decomposition strategies on optimization performance. Therefore, all decomposition-based optimization methods are provided with precise structural details, and each utilizes 3×10^6 FEs for optimization. When comparing RLDO with non-decomposition-based methods, the computational resources consumed by structural identification in RLDO are deducted from the total 3×10^6 FEs to ensure a fair comparison. The population size for all algorithms is set to 50. The optimization results are obtained through 25 independent runs. A Wilcoxon rank-sum test is conducted on the optimization results at a significance level of 0.05 to compare the proposed RLDO with the other algorithms.

D. Optimization Results On Conflicting Benchmark

The experimental results of RLDO in comparison with decomposition-based optimization methods on large-scale conflicting overlapping problems with complex topologies are presented in Table I. Additionally, the convergence curves for f_1 , f_2 , and f_6 are shown in Fig. 3. Due to the page limit,

the remaining convergence curves are presented in Section S.I of the Supplementary Material. The results comparing RLDO with non-decomposition-based optimization methods are presented in Table II. The performance of RLDO is evaluated in comparison with various optimization methods, including sequential decomposition, random decomposition, and non-decomposition-based approaches.

- 1) Compared to sequential decomposition-based optimization methods ORDG and CBCCO, RLDO demonstrates superior performance on more than 13 benchmark functions while showing inferior results on only one function. Both ORDG and CBCCO adopt the traditional CC framework to optimize each subcomponent of the LSOP sequentially. CBCCO improves upon ORDG by allocating shared variables to the subcomponent with the largest contribution. However, optimizing each subcomponent individually may lead to less efficient optimization performance. ORDG shares variables across neighboring subcomponents, which may introduce additional optimization conflicts. Furthermore, due to the complexity of the interaction relationship in LSOPs with complex topologies, the shared variable allocation strategy of CBCCO fails to guarantee improved global optimization performance.
- 2) Compared to RDG3, RLDO performs significantly better on at least 11 benchmark problems, but is inferior on 3 problems. RDG3 divides the overlapping problem into multiple independent variable groups by setting a threshold of group size. However, the threshold selection poses a significant challenge for RDG3, as different thresholds can lead to significant differences in optimization performance. Additionally, the decomposition results of RDG3 are influenced by the initially detected variables. Specifically, different initially detected variables may lead to varying decomposition results, even when the threshold is the same. Moreover, RDG3 focuses solely on the subcomponent size for decomposition, overlooking the critical feature of subcomponent weights.
- 3) Compared with Random, RLDO outperforms it on 14 functions, performs equally well on 1 function, and shows inferior optimization performance on 1 problems. Random breaks the linkage between subcomponents to divide the overlapping problem into multiple subcomponents. Compared to RDG3, Random method decomposes the problem from a global perspective, allowing it to capture the closer interaction between multiple subcomponents in LSOPs with complex topologies. However, due to its inherent randomness, Random struggles to fully explore the key features of overlapping problems, failing to achieve rational and efficient decomposition.
- 4) Compared to non-decomposition-based methods (SaNSDE, CSO, and MMES), RLDO demonstrates superior performance on at least 10 problems and inferior performance on no more than 5 problems. Non-decomposition-based methods optimize the overlapping problem as a whole, while neglecting to utilize the

potential modular structure characteristics. The high dimensionality of such problems results in a significant expansion of the search space, increasing the search difficulty for the non-decomposition-based methods. Furthermore, these methods fail to extract the key components of the problem, further limiting their optimization efficiency. It is observed from Table II that RLDO exhibits weaker performance than MMES on 5 out of the 16 benchmark problems. First, MMES is one of the leading algorithms in the field of LSGO, and it inherently possesses strong capabilities for handling high-dimensional problems. Second, the decomposition model in RLDO is trained using a DE-based optimizer, which results in a reduced adaptability of the RLDO decomposition model to the MMES algorithm, compared to its compatibility with SaNSDE. Nonetheless, RLDO consistently outperforms all non-decomposition-based global optimization algorithms, reflecting that a rational decomposition strategy can significantly enhance the optimization efficiency of LSOPs.

Overall, the results indicate that RLDO ranks among the best of the compared algorithms. Unlike traditional sequential and random decomposition approaches, RLDO utilizes a learning-based mechanism to acquire comprehensive knowledge about the decomposition of LSOP during the training process. By learning the optimization feedback information from a global perspective, RLDO develops an efficient decomposition strategy for LSOP with various topological structures. Through its well-trained policy network, RLDO is capable of analyzing complex problem characteristics and performing effective decomposition, resulting in superior optimization performance.

Nevertheless, it is worth noting that RLDO exhibits suboptimal performance on certain problems such as f_{16} . This limitation can be attributed to two primary factors. First, RLDO is trained on a limited set of LSOPs, which may restrict its ability to generalize to problems that exhibit significantly different topological structures, degrees of overlap, or subcomponent weight distributions. Second, RLDO adopts a basic cooperative co-evolution (CC) framework with a standard differential evolution (DE) optimizer. This simplified optimization setting may hinder the accurate assessment of decomposition quality, potentially resulting in imperfect or misleading reward signals during training.

E. Optimization Results on Conforming Benchmark

The experimental results of RLDO in comparison with decomposition-based optimization methods on conforming overlapping problems are presented in Table III. Due to the page limit, the convergence curves are presented in Section S.II of the Supplementary Material. The results comparing RLDO with non-decomposition-based optimization methods are presented in Table IV. RLDO also demonstrates superior optimization results compared to other algorithms on conforming overlapping benchmark problems.

- 1) Compared to sequential decomposition-based optimization methods ORDG and CBCCO, RLDO outperforms

TABLE I

THE OPTIMIZATION RESULTS OF VARIOUS DECOMPOSITION-BASED METHODS ON LARGE-SCALE COMPLEX TOPOLOGICAL CONFLICTING OVERLAPPING BENCHMARK. THE SYMBOLS “W”, “T”, AND “L” INDICATE THAT RLDO PERFORMS SIGNIFICANTLY BETTER THAN, SIMILAR TO, AND WORSE THAN THE CORRESPONDING ALGORITHM BY WILCOXON’S RANK-SUM TEST AT THE SIGNIFICANCE OF 0.05, RESPECTIVELY.

Func.	ORDG	CBCCO	RDG3-I	RDG3-II	Random	RLDO
f_1	2.96E+11 ± 5.28E+10(+)	3.46E+11 ± 8.47E+10(+)	3.76E+11 ± 5.68E+10(+)	5.85E+11 ± 9.61E+10(+)	3.98E+11 ± 5.22E+10(+)	1.44E+11 ± 1.80E+10
f_2	2.68E+16 ± 7.39E+15(+)	3.71E+16 ± 1.01E+16(+)	1.06E+16 ± 2.99E+15(+)	7.53E+15 ± 2.15E+15(≈)	7.81E+15 ± 1.86E+15(+)	6.64E+15 ± 2.05E+15
f_3	8.98E+11 ± 2.44E+11(+)	8.58E+11 ± 2.23E+11(+)	2.20E+11 ± 4.98E+10(-)	4.56E+11 ± 7.97E+10(-)	2.18E+12 ± 5.60E+11(+)	8.08E+11 ± 1.91E+11
f_4	3.01E+10 ± 6.92E+09(≈)	4.43E+10 ± 1.17E+10(+)	5.05E+10 ± 9.98E+09(+)	5.01E+10 ± 7.79E+09(+)	4.36E+10 ± 7.96E+09(+)	3.54E+10 ± 5.22E+09
f_5	1.26E+10 ± 2.36E+09(≈)	1.33E+10 ± 2.23E+09(≈)	1.79E+10 ± 2.61E+09(+)	1.81E+10 ± 2.80E+09(+)	1.98E+10 ± 3.58E+09(+)	1.31E+10 ± 1.27E+09
f_6	3.62E+10 ± 7.30E+09(+)	3.89E+10 ± 1.11E+10(+)	4.43E+10 ± 7.38E+09(+)	6.14E+10 ± 9.77E+09(+)	4.34E+10 ± 8.67E+09(+)	1.74E+10 ± 2.81E+09
f_7	4.09E+10 ± 8.51E+09(+)	3.33E+10 ± 6.19E+09(+)	4.92E+10 ± 8.05E+09(+)	6.54E+10 ± 1.30E+10(+)	4.73E+10 ± 1.21E+10(+)	2.23E+10 ± 5.14E+09
f_8	3.34E+12 ± 6.71E+11(-)	4.28E+12 ± 8.97E+11(-)	3.84E+12 ± 7.08E+11(-)	4.88E+12 ± 8.14E+11(≈)	8.48E+12 ± 1.86E+12(+)	5.13E+12 ± 9.64E+11
f_9	8.88E+10 ± 8.97E+09(+)	9.49E+10 ± 1.67E+10(+)	9.03E+10 ± 1.07E+10(+)	9.55E+10 ± 1.41E+10(+)	8.75E+10 ± 1.32E+10(+)	4.65E+10 ± 6.10E+09
f_{10}	6.43E+10 ± 1.04E+10(+)	7.72E+10 ± 1.71E+10(+)	6.61E+10 ± 1.08E+10(+)	2.90E+10 ± 8.45E+09(≈)	5.97E+10 ± 1.23E+10(+)	2.63E+10 ± 4.77E+09
f_{11}	2.96E+08 ± 6.97E+07(+)	4.05E+08 ± 9.95E+07(+)	3.93E+08 ± 1.03E+08(+)	4.34E+08 ± 6.23E+07(+)	5.32E+08 ± 8.73E+07(+)	1.34E+08 ± 2.46E+07
f_{12}	8.57E+09 ± 1.22E+09(+)	1.31E+10 ± 2.52E+09(+)	1.19E+10 ± 1.41E+09(+)	1.53E+10 ± 3.87E+09(+)	2.00E+10 ± 3.88E+09(+)	6.65E+09 ± 1.04E+09
f_{13}	3.52E+12 ± 1.00E+12(+)	4.23E+12 ± 8.49E+11(+)	2.28E+12 ± 4.04E+11(+)	5.04E+12 ± 7.86E+11(+)	1.88E+12 ± 2.81E+11(≈)	1.84E+12 ± 3.17E+11
f_{14}	7.25E+11 ± 7.78E+10(+)	8.27E+11 ± 8.28E+10(+)	6.30E+11 ± 7.32E+10(+)	4.24E+11 ± 7.30E+10(+)	4.50E+11 ± 5.38E+10(+)	3.45E+11 ± 4.32E+10
f_{15}	2.55E+09 ± 3.83E+08(+)	4.07E+09 ± 8.36E+08(+)	3.13E+09 ± 4.10E+08(+)	4.60E+09 ± 5.75E+08(+)	2.62E+09 ± 2.91E+08(+)	1.87E+09 ± 2.44E+08
f_{16}	5.73E+13 ± 1.69E+13(+)	8.69E+13 ± 2.40E+13(+)	1.55E+13 ± 3.96E+12(-)	3.01E+13 ± 7.05E+12(-)	3.13E+13 ± 4.59E+12(-)	4.56E+13 ± 6.79E+12
W/T/L	13/2/1	14/1/1	13/0/3	11/3/2	14/1/1	

TABLE II

THE OPTIMIZATION RESULTS OF NON-DECOMPOSITION-BASED METHODS AND THEIR RLDO VARIANTS ON LARGE-SCALE COMPLEX TOPOLOGICAL CONFLICTING OVERLAPPING BENCHMARK. THE SYMBOLS “W”, “T”, AND “L” INDICATE THAT RLDO PERFORMS SIGNIFICANTLY BETTER THAN, SIMILAR TO, AND WORSE THAN THE CORRESPONDING ALGORITHM BY WILCOXON’S RANK-SUM TEST AT THE SIGNIFICANCE OF 0.05, RESPECTIVELY.

Func.	SaNSDE	RLDO-SaNSDE	CSO	RLDO-CSO	MMES	RLDO-MMES
f_1	3.49E+11 ± 3.60E+10(+)	1.45E+11 ± 1.78E+10	3.18E+11 ± 2.82E+10(+)	1.90E+11 ± 2.78E+10	7.13E+09 ± 9.49E+08(+)	4.68E+09 ± 6.12E+08
f_2	1.45E+16 ± 4.09E+15(+)	6.71E+15 ± 1.98E+15	1.06E+16 ± 1.29E+15(-)	1.45E+16 ± 2.02E+15	1.08E+13 ± 2.23E+12(-)	4.52E+13 ± 4.86E+12
f_3	6.84E+11 ± 1.74E+11(-)	8.22E+11 ± 1.90E+11	9.79E+10 ± 1.52E+09(-)	1.99E+11 ± 3.75E+10	6.91E+09 ± 5.18E+08(+)	1.23E+09 ± 3.01E+08
f_4	3.65E+10 ± 5.06E+09(≈)	3.57E+10 ± 4.17E+09	4.02E+10 ± 4.31E+09(+)	2.46E+10 ± 4.88E+09	5.44E+08 ± 5.20E+07(+)	3.72E+08 ± 5.58E+07
f_5	2.71E+10 ± 3.77E+09(+)	1.33E+10 ± 1.26E+09	2.25E+10 ± 1.92E+09(+)	9.54E+09 ± 1.16E+09	1.43E+08 ± 1.58E+07(-)	6.54E+08 ± 8.17E+07
f_6	3.26E+10 ± 4.25E+09(+)	1.76E+10 ± 2.79E+09	3.21E+10 ± 2.66E+09(+)	1.78E+10 ± 2.03E+09	5.34E+08 ± 4.67E+07(+)	2.54E+08 ± 2.99E+07
f_7	5.38E+10 ± 9.88E+09(+)	2.25E+10 ± 5.08E+09	4.93E+10 ± 4.30E+09(+)	2.94E+10 ± 8.61E+09	6.73E+08 ± 7.21E+07(+)	1.41E+08 ± 3.15E+07
f_8	5.32E+12 ± 7.65E+11(≈)	5.33E+12 ± 1.00E+12	5.10E+12 ± 3.44E+11(+)	2.37E+12 ± 6.77E+11	7.48E+10 ± 6.26E+09(-)	4.21E+11 ± 1.09E+11
f_9	7.83E+10 ± 6.58E+09(+)	4.69E+10 ± 6.05E+09	6.96E+10 ± 5.35E+09(+)	3.98E+10 ± 4.78E+09	8.75E+09 ± 6.32E+08(+)	3.21E+09 ± 5.88E+08
f_{10}	6.39E+10 ± 1.09E+10(+)	2.65E+10 ± 4.69E+09	7.20E+10 ± 8.69E+09(+)	3.99E+10 ± 7.20E+09	9.70E+08 ± 1.23E+08(+)	6.22E+08 ± 1.09E+08
f_{11}	4.39E+08 ± 5.72E+07(+)	1.35E+08 ± 2.43E+07	3.79E+08 ± 5.05E+07(-)	5.50E+08 ± 8.71E+07	5.58E+06 ± 4.73E+05(≈)	5.49E+06 ± 9.64E+05
f_{12}	2.18E+10 ± 2.50E+09(+)	6.72E+09 ± 1.03E+09	2.03E+10 ± 9.08E+08(+)	7.49E+09 ± 1.53E+09	9.77E+07 ± 8.88E+06(+)	7.81E+07 ± 1.50E+07
f_{13}	2.44E+12 ± 3.43E+11(+)	1.85E+12 ± 3.15E+11	2.64E+12 ± 2.67E+11(≈)	2.52E+12 ± 5.30E+10	4.31E+10 ± 2.81E+09(-)	1.46E+11 ± 3.04E+10
f_{14}	5.14E+11 ± 8.34E+10(+)	3.47E+11 ± 4.26E+10	5.13E+11 ± 6.83E+10(≈)	5.23E+12 ± 6.60E+11	6.50E+10 ± 5.38E+10(+)	2.76E+10 ± 4.24E+09
f_{15}	3.60E+09 ± 3.97E+08(+)	1.89E+09 ± 2.41E+08	3.17E+09 ± 3.39E+08(+)	1.86E+09 ± 2.98E+08	7.75E+07 ± 9.91E+06(+)	3.28E+07 ± 4.68E+06
f_{16}	2.92E+13 ± 4.46E+12(-)	4.58E+13 ± 6.74E+12	4.01E+13 ± 2.67E+12(+)	2.89E+13 ± 5.49E+12	5.21E+11 ± 5.53E+10(-)	9.96E+11 ± 2.96E+11
W/T/L	12/2/2		11/2/3		10/1/5	

* To ensure experimental fairness, all 3×10^6 FEs in RLDO’s variants are utilized to include both the LSOPs structures identification and the optimization process.

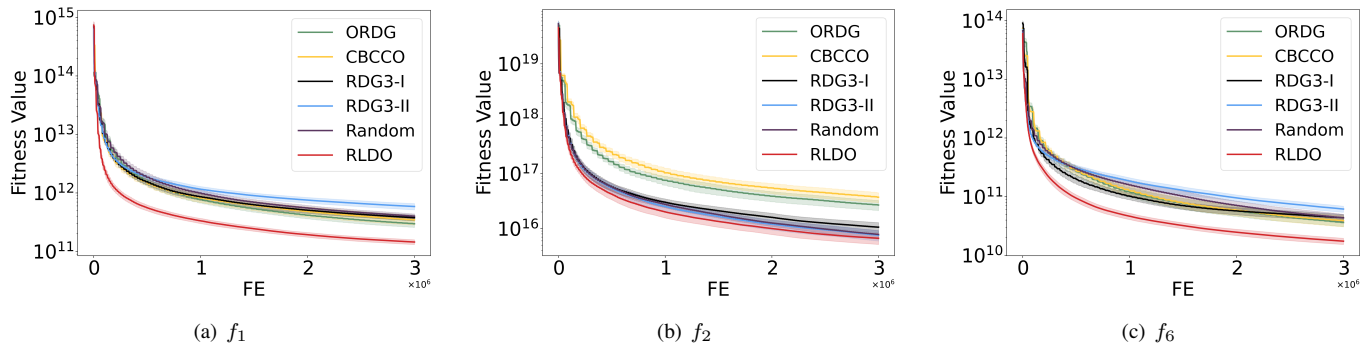


Fig. 3: Convergence curves for RLDO and its comparison algorithms to solve the f_1 , f_2 , and f_6 in large-scale complex topological overlapping benchmark.

TABLE III

THE OPTIMIZATION RESULTS OF VARIOUS DECOMPOSITION-BASED METHODS ON LARGE-SCALE COMPLEX TOPOLOGICAL CONFORMING OVERLAPPING BENCHMARK. THE SYMBOLS “W”, “T”, AND “L” INDICATE THAT RLDO PERFORMS SIGNIFICANTLY BETTER THAN, SIMILAR TO, AND WORSE THAN THE CORRESPONDING ALGORITHM BY WILCOXON’S RANK-SUM TEST AT THE SIGNIFICANCE OF 0.05, RESPECTIVELY.

Func.	ORDG	CBCCO	RDG3-I	RDG3-II	Random	RLDO
f_1	5.45E+09 ± 1.88E+09(+)	6.31E+09 ± 1.95E+09(+)	5.96E+09 ± 9.91E+08(+)	5.62E+09 ± 9.79E+08(+)	1.32E+10 ± 2.06E+09(+)	2.93E+09 ± 3.18E+08
f_2	7.10E+09 ± 1.06E+09(+)	1.10E+10 ± 2.17E+09(+)	7.87E+09 ± 8.66E+08(+)	7.44E+09 ± 1.73E+09(+)	1.26E+10 ± 1.56E+09(+)	4.06E+09 ± 4.90E+08
f_3	3.25E+11 ± 6.10E+10(+)	4.64E+11 ± 1.03E+11(+)	2.59E+11 ± 5.84E+10(+)	3.06E+11 ± 6.33E+10(+)	2.59E+11 ± 4.35E+10(+)	2.00E+11 ± 4.97E+10
f_4	6.81E+12 ± 1.45E+12(+)	8.20E+12 ± 1.78E+12(+)	1.94E+12 ± 3.49E+11(−)	2.92E+12 ± 3.57E+11(≈)	1.87E+12 ± 4.53E+11(−)	2.79E+12 ± 6.61E+11
f_5	4.46E+09 ± 9.14E+08(+)	6.38E+09 ± 1.45E+09(+)	3.16E+09 ± 5.31E+08(+)	4.05E+09 ± 6.83E+08(+)	3.50E+09 ± 4.87E+08(+)	1.81E+09 ± 2.56E+08
f_6	4.57E+11 ± 2.95E+10(+)	5.18E+11 ± 7.50E+10(+)	5.78E+11 ± 6.52E+10(+)	7.41E+11 ± 1.11E+11(+)	4.74E+11 ± 5.02E+10(+)	2.23E+11 ± 2.60E+10
f_7	5.83E+11 ± 8.90E+10(+)	3.83E+11 ± 6.05E+10(+)	4.91E+11 ± 1.07E+11(+)	1.03E+12 ± 1.66E+11(+)	4.49E+11 ± 7.51E+10(+)	2.70E+11 ± 4.80E+10
f_8	9.35E+11 ± 3.22E+11(+)	1.52E+12 ± 4.44E+11(+)	6.67E+11 ± 9.81E+10(+)	7.05E+11 ± 9.50E+10(+)	6.30E+11 ± 1.54E+11(+)	1.68E+11 ± 3.05E+10
f_9	1.97E+12 ± 4.08E+11(≈)	2.57E+12 ± 9.02E+11(≈)	1.80E+12 ± 2.20E+11(−)	2.44E+12 ± 4.27E+11(≈)	2.02E+12 ± 2.07E+11(≈)	2.11E+12 ± 3.36E+11
f_{10}	8.75E+13 ± 3.86E+13(+)	9.40E+13 ± 3.17E+13(+)	1.70E+13 ± 3.28E+12(−)	4.28E+13 ± 1.14E+13(+)	2.56E+13 ± 4.74E+12(≈)	2.94E+13 ± 4.47E+12
f_{11}	1.95E+14 ± 5.71E+13(+)	7.87E+13 ± 2.50E+13(+)	3.76E+13 ± 7.42E+12(+)	4.41E+13 ± 1.20E+13(+)	4.34E+13 ± 1.01E+13(+)	2.43E+13 ± 7.05E+12
f_{12}	7.50E+09 ± 1.77E+09(+)	7.19E+09 ± 1.59E+09(+)	6.30E+09 ± 9.13E+08(+)	1.11E+10 ± 2.84E+09(+)	1.18E+10 ± 2.46E+09(+)	2.51E+09 ± 6.75E+08
f_{13}	9.42E+11 ± 1.78E+11(+)	9.80E+11 ± 2.81E+11(+)	2.25E+12 ± 4.64E+11(+)	1.34E+12 ± 2.40E+11(+)	1.78E+12 ± 5.09E+11(+)	3.02E+11 ± 4.97E+10
f_{14}	2.56E+11 ± 6.64E+10(+)	2.65E+11 ± 4.51E+10(+)	5.54E+11 ± 1.06E+11(+)	4.71E+11 ± 4.06E+10(+)	1.61E+11 ± 3.10E+10(≈)	1.71E+11 ± 3.13E+10
f_{15}	2.43E+11 ± 6.26E+10(+)	2.60E+11 ± 3.83E+10(+)	1.38E+11 ± 1.65E+10(+)	6.91E+11 ± 1.74E+11(+)	1.46E+11 ± 2.37E+10(+)	8.88E+10 ± 1.53E+10
f_{16}	3.48E+11 ± 8.95E+10(≈)	2.53E+11 ± 2.55E+10(−)	4.13E+11 ± 7.54E+10(≈)	4.59E+11 ± 8.23E+10(≈)	2.87E+11 ± 7.33E+10(−)	4.33E+11 ± 7.29E+10
W/T/L	14/2/0	14/1/1	12/1/3	13/3/0	11/3/2	

TABLE IV

THE OPTIMIZATION RESULTS OF NON-DECOMPOSITION-BASED METHODS AND THEIR RLDO VARIANTS ON LARGE-SCALE COMPLEX TOPOLOGICAL CONFLICTING OVERLAPPING BENCHMARK. THE SYMBOLS “W”, “T”, AND “L” INDICATE THAT RLDO PERFORMS SIGNIFICANTLY BETTER THAN, SIMILAR TO, AND WORSE THAN THE CORRESPONDING ALGORITHM BY WILCOXON’S RANK-SUM TEST AT THE SIGNIFICANCE OF 0.05, RESPECTIVELY.

Func.	SanSDE	RLDO-SanSDE	CSO	RLDO-CSO	MMES	RLDO-MMES
f_1	7.73E+09 ± 7.37E+08(+)	2.95E+09 ± 3.14E+08	5.82E+09 ± 4.56E+08(+)	3.43E+09 ± 7.40E+08	5.52E+07 ± 5.22E+06(+)	1.82E+07 ± 2.08E+06
f_2	8.07E+09 ± 1.92E+09(+)	4.10E+09 ± 4.82E+08	7.10E+09 ± 4.77E+08(+)	3.70E+09 ± 6.39E+08	6.59E+08 ± 6.86E+07(+)	2.53E+08 ± 5.31E+07
f_3	1.87E+11 ± 2.78E+10(≈)	2.01E+11 ± 4.92E+10	2.14E+11 ± 2.97E+10(+)	1.76E+11 ± 4.27E+10	3.18E+09 ± 4.60E+08(+)	1.18E+09 ± 2.72E+08
f_4	3.57E+12 ± 7.96E+11(+)	2.81E+12 ± 6.55E+11	3.99E+12 ± 5.42E+11(≈)	3.80E+12 ± 1.25E+12	4.96E+10 ± 3.06E+09(−)	8.01E+10 ± 1.58E+10
f_5	5.03E+09 ± 7.00E+08(+)	1.87E+09 ± 2.57E+08	5.75E+09 ± 6.53E+08(+)	3.14E+09 ± 6.86E+08	5.18E+07 ± 8.58E+06(+)	2.41E+07 ± 3.86E+06
f_6	4.95E+11 ± 8.87E+10(+)	2.25E+11 ± 2.58E+10	4.49E+11 ± 4.40E+10(+)	2.22E+11 ± 3.88E+10	2.34E+09 ± 2.67E+08(+)	8.08E+08 ± 1.99E+08
f_7	4.05E+11 ± 5.14E+10(+)	2.72E+11 ± 4.73E+10	5.06E+11 ± 6.00E+10(+)	3.26E+11 ± 6.77E+10	5.51E+09 ± 6.21E+08(+)	2.63E+09 ± 5.15E+08
f_8	5.98E+11 ± 1.14E+11(+)	1.70E+11 ± 3.02E+10	2.85E+11 ± 5.68E+10(−)	3.86E+11 ± 9.65E+10	7.29E+09 ± 8.54E+08(−)	5.33E+10 ± 1.02E+10
f_9	1.70E+12 ± 2.65E+11(−)	2.13E+12 ± 3.32E+11	2.05E+12 ± 1.79E+11(−)	3.45E+12 ± 4.88E+11	6.52E+09 ± 8.32E+08(−)	1.16E+10 ± 2.62E+09
f_{10}	2.11E+13 ± 2.47E+12(−)	2.97E+13 ± 4.42E+12	4.51E+13 ± 4.77E+12(−)	6.43E+13 ± 1.29E+13	2.25E+11 ± 3.57E+10(−)	4.75E+11 ± 6.19E+10
f_{11}	3.78E+13 ± 7.70E+12(+)	2.45E+13 ± 6.98E+12	3.49E+13 ± 4.71E+12(≈)	3.41E+13 ± 7.49E+12	3.61E+11 ± 4.73E+10(+)	1.15E+11 ± 2.02E+10
f_{12}	6.01E+09 ± 1.07E+09(+)	2.54E+09 ± 6.69E+08	7.22E+09 ± 9.24E+08(+)	3.03E+09 ± 5.78E+08	6.00E+07 ± 7.88E+06(+)	2.45E+07 ± 6.04E+06
f_{13}	8.42E+11 ± 1.07E+11(+)	3.05E+11 ± 4.90E+10	9.36E+11 ± 1.19E+11(+)	3.84E+11 ± 9.18E+10	2.62E+09 ± 2.81E+08(+)	8.46E+08 ± 2.04E+08
f_{14}	3.41E+11 ± 6.16E+10(+)	1.74E+11 ± 3.09E+10	3.10E+11 ± 2.88E+10(+)	1.68E+11 ± 3.11E+10	3.50E+09 ± 4.38E+08(−)	7.56E+09 ± 1.24E+09
f_{15}	2.39E+11 ± 3.58E+10(+)	8.97E+10 ± 1.51E+10	3.04E+11 ± 1.86E+10(+)	1.32E+11 ± 2.95E+10	2.66E+09 ± 3.46E+08(+)	8.04E+08 ± 1.69E+08
f_{16}	5.91E+11 ± 4.39E+10(+)	4.37E+11 ± 9.25E+10	2.94E+11 ± 4.03E+10(≈)	2.74E+11 ± 4.51E+10	5.40E+09 ± 6.59E+08(+)	1.33E+09 ± 2.89E+08
W/T/L	13/1/2		10/3/3		11/0/5	

* To ensure experimental fairness, all 3×10^6 FEs in RLDO’s variants are utilized to include both the LSOPs structures identification and the optimization process.

them on 14 benchmark functions, while showing inferior results on 1 functions. In the conforming overlapping problems, the shared variables have the same optimal values in adjacent subcomponents. Consequently, when optimizing a certain subcomponent in such problems, the optimized shared variables may contribute to the optimization of the adjacent subcomponents. It can be observed that ORDG outperforms CBCCO in the sequential decomposition method, as sharing variables across multiple subcomponents enhances optimization efficiency. However, RLDO provides a reasonable decomposition of the overlapping problem from a global perspective, yielding significantly better results than both ORDG and CBCCO.

- 2) Compared to RDG3-I, RLDO outperforms it on 12 problems, achieves comparable results on 1 problems, and performs worse on 3 problem. Compared to RDG3-II, RLDO outperforms it on 13 problems and performs equally well on 3 problems. The smaller subcomponent size of RDG3-I allows for better collaboration between subcomponents during the CC optimization process, resulting in better optimization results compared to RDG3-

II. However, by decomposing the problem solely based on the size of the subcomponents, RDG3 fails to achieve better optimization results than RLDO.

- 3) Compared to Random, RLDO outperforms it on 11 functions but shows inferior optimization performance on 2 problems. In contrast to conflicting overlapping problems, Random achieves better optimization performance on the conforming overlapping problems. This is because the conflict between subcomponents resulting from the decomposition of Random is smaller in conforming overlapping problems than in conflicting ones. However, due to the inherent randomness of the Random method, its optimization performance is significantly less effective than that of RLDO.
- 4) Compared to non-decomposition-based methods, RLDO performs better on at least 10 problems and worse on no more than 5 problems. By learning problem-specific decompositions, RLDO achieves better results than the corresponding global optimizers, further demonstrating its outstanding performance in handling large-scale overlapping problems. Moreover, this further emphasizes the necessity of achieving reasonable decomposition for

optimizing LSOPs.

F. Analysis of the decomposition results

To investigate how decomposition impacts optimization performance, we conducted a comparative analysis of the decomposition mechanisms used by RLDO and its comparison algorithms.

We selected two representative functions, f_5 and f_{13} , from the large-scale large-scale conflicting overlapping benchmark set to illustrate the decomposition and optimization performance of each algorithm. The original problems f_5 and f_{13} , along with their corresponding decomposition results generated by each algorithm, are shown in Fig. 4. The optimization results for f_5 and f_{13} obtained by each algorithm can be found in Table I.

The original subcomponent interaction matrices and their corresponding topological graphs for f_5 and f_{13} are shown on the left side of Figs. 4(a) and 4(b), respectively. In each subcomponent interaction matrix M , the color intensity at position M_{ij} reflects the interaction strength between subcomponents i and j , which is determined by the number of shared variables. The topological graphs below matrix M visually illustrate the interaction network, where larger nodes represent subcomponents with greater weight. On the right side of Figs. 4(a) and 4(b), we present the decomposition results of each algorithm. A dark blue cell at W_{ij} indicates that the linkage between subcomponents i and j has been preserved in the decomposition.

On problem f_5 , both ORDG and CBCCO achieved the best optimization performance, whereas Random yielded the best result on f_{13} , as presented in Table I. These outcomes suggest that the optimal decomposition strategy varies significantly depending on the characteristics of the problem—such as topological structure, overlapping degree, and subcomponent weight. Remarkably, RLDO achieved the best optimization performance on both f_5 and f_{13} , demonstrating its ability to adaptively discover effective decomposition strategies.

Further inspection of Fig. 4(b) reveals that RLDO's decomposition for f_5 closely resembles that of the sequential decomposition approach, where the problem is divided into many smaller subcomponents. Similarly, for f_{13} , RLDO's decomposition results align closely with those of the Random method. These similarities indicate that RLDO is capable of analyzing the unique characteristics of each problem and adjusting its decomposition strategy accordingly to maximize optimization performance.

RLDO accomplishes this adaptability through its interactive learning mechanism. During training, it collects reward and penalty signals from different decomposition attempts and uses this feedback to iteratively refine its policy network. This process enables the algorithm to extract the essential features of each problem and derive an optimal decomposition strategy tailored to the problem's structure.

In contrast, other algorithms are constrained by static decomposition rules or random mechanisms. CBCCO, ORDG, and SaNSDE, for instance, yield identical decomposition structures across all test problems, indicating their inability

to adjust to different problem types. RDG3 attempts to merge highly interactive subcomponents based on a predefined subcomponent size threshold.

For RDG3-II, the larger threshold leads to fewer but larger groups, while RDG3-I results in more, smaller subcomponents. As shown in Fig. 4, RDG3-II exhibits more inter-group interactions, but relying solely on subcomponent size fails to ensure effective decomposition. Moreover, the initial selection of detected variables significantly affects RDG3's results, introducing additional randomness.

The Random method decomposes the problem by probabilistically breaking links between subcomponents, but its effectiveness is hindered by the variability introduced through different disconnection probabilities. As a result, it often lacks consistency in producing meaningful decompositions.

Both ORDG and CBCCO eliminate all links between subcomponents in LSOP, resulting in empty decomposition matrices, as depicted in Fig. 4. This strategy typically fails for overlapping problems, as optimizing each subcomponent in isolation may lead to excessive reliance on shared variables and reduced overall efficiency. Although CBCCO attempts to mitigate this by assigning shared variables to the most influential subcomponent, and ORDG by distributing them across adjacent groups, the similarity in their performance suggests that such allocation strategies have limited impact on overall optimization outcomes.

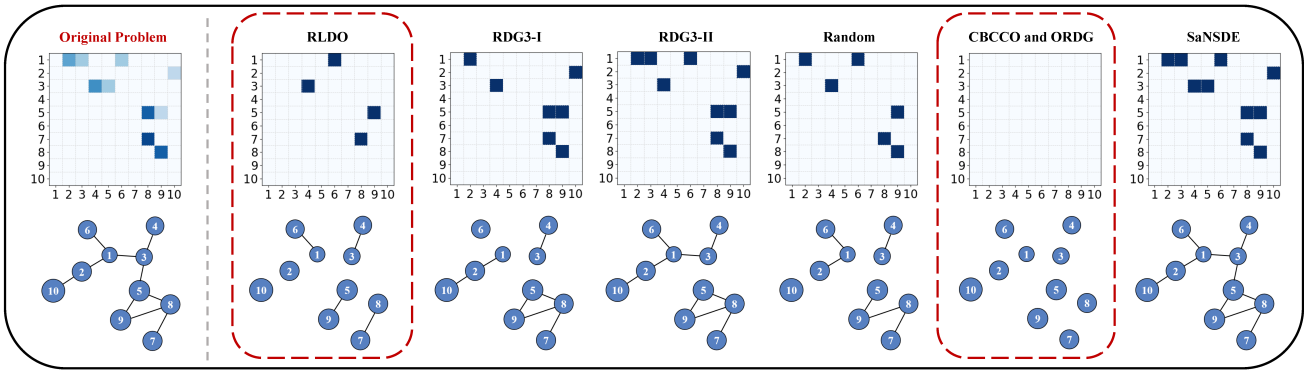
Unlike these methods, SaNSDE treats the problem as a whole, resulting in no decomposition. Its decomposition matrix is identical to the original subcomponent interaction matrix, and it fails to reduce dimensionality or exploit modular structures. Experimental results confirm that large-scale overlapping problems benefit significantly from well-structured decompositions, reinforcing the need for adaptive and learning-based decomposition strategies like RLDO.

It is important to note that RLDO underperforms in certain problem instances. As shown in Tables I and III, RLDO fails to achieve the best performance on ten problems across the two benchmarks. The primary reasons are the insufficient isolation of high-weight subcomponents and imbalanced sizes of the optimized subgroups. To illustrate this, we selected two representative problems, f_3 and f_8 , from the conflicting overlapping benchmark set. Due to the page limit, the detailed decomposition results are provided in Section S.III of the Supplementary Material.

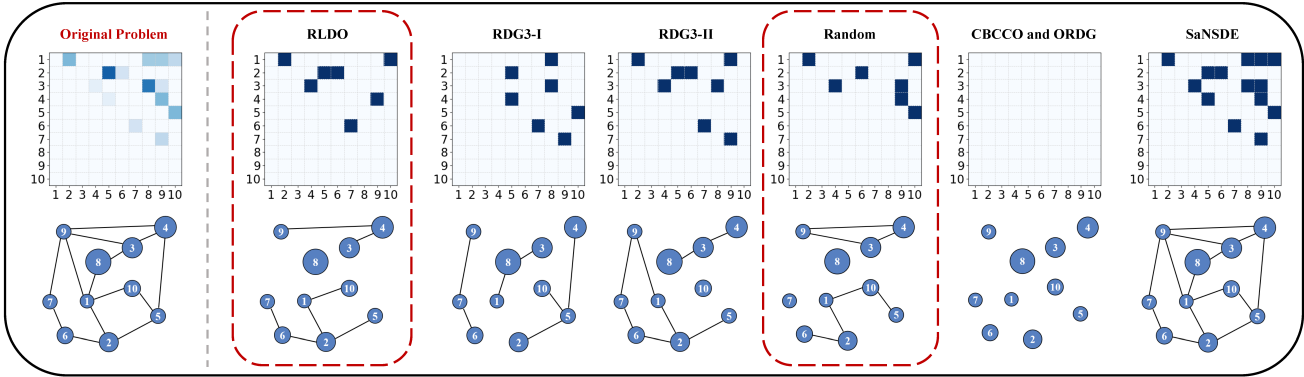
In both f_3 and f_8 , RLDO fails to form relatively independent optimized subgroups for the subcomponents with larger weights, resulting in insufficient focus on the important subgroups during the CC optimization process. Moreover, the strong size imbalance among subgroups in f_3 and f_8 leads to increased optimization conflicts and inefficient allocation of computational resources, further hindering the overall optimization performance.

G. Optimization Results On Problems of Different Scales

To evaluate the generalization ability and scalability of RLDO, we evaluated its performance on a series of overlapping problems with nominal dimensionalities of 200, 500,



(a) The large-scale conflicting overlapping benchmark problem f_5 and the decomposition results of each algorithm.



(b) The large-scale conflicting overlapping benchmark problem f_{13} and the decomposition results of each algorithm.

Fig. 4: The original problems and the corresponding decomposition results for each algorithm, represented by subcomponent interaction matrices and topological graphs.

1000, 2000, and 3000. Due to the use of shared variables across overlapping components, the actual number of unique decision variables is slightly smaller in each case (e.g., effective dimensions around 170 – 2800). These problems were constructed using the CTOC framework by varying the subcomponent size ($s = 20, 50, 100, 200, 300$). For each scale, the overlapping degree was set to one-tenth of the corresponding subcomponent size. RDG3-I and RDG3-II represent the RDG3 method with subcomponent size thresholds of $0.2 \times s$ and $0.4 \times s$, respectively.

Due to the page limit, the detailed optimization results are presented in Section S.III of the Supplementary Material. The Wilcoxon's rank-sum test results for RLDO and its comparison algorithms at the significance level of 0.05 are shown in Table V.

- 1) Compared to sequential decomposition-based optimization methods (ORDG and CBCCO), RLDO consistently outperforms ORDG and CBCCO across all problem scales. These sequential methods optimize only a limited subset of variables per step and suffer from poor optimization efficiency, especially in small-scale problems.
- 2) Compared to other decomposition strategies (RDG3-I, RDG3-II, and Random), RLDO achieves superior performance across all scales by leveraging a learning mechanism to acquire and apply effective decomposition knowledge, rather than relying on fixed heuristics.

TABLE V
THE OPTIMIZATION RESULTS OF VARIOUS METHODS ON OVERLAPPING PROBLEMS OF DIFFERENT SCALES, WHERE EACH ENTRY INDICATES THE NUMBER OF PROBLEMS IN WHICH RLDO PERFORMS BETTER THAN, SIMILAR TO, OR WORSE THAN THE CORRESPONDING METHOD, RESPECTIVELY. n REPRESENTS THE NUMBER OF SUBGROUPS, WITH A VALUE OF 10, AND s REPRESENTS THE SUBGROUP SIZE, WHICH TAKES THE VALUES 20, 50, 100, 200, AND 300.

$n \times s$	ORDG	CBCCO	RDG3-I	RDG3-II	Random	SaNSDE	RLDO
200	8/0/0	8/0/0	5/2/1	5/3/0	6/2/0	2/3/3	-
500	7/1/0	8/0/0	6/1/1	7/1/0	7/1/0	7/0/1	-
1000	6/0/2	6/0/2	6/2/0	7/1/0	7/0/1	6/1/1	-
2000	7/0/1	7/1/0	6/1/1	6/2/0	7/0/1	8/0/0	-
3000	6/1/1	7/0/1	8/0/0	7/1/0	6/2/0	8/0/0	-

- 3) Compared to non-decomposition-based optimization method (SaNSDE), on the smallest-scale problems, SaNSDE performs comparably or slightly better than RLDO due to the minimal scalability burden. However, as problem dimensionality increases, RLDO significantly outperforms SaNSDE, whose performance deteriorates under high-dimensional settings.

V. CONCLUSION

In this paper, we propose a novel reinforcement learning-based framework, termed RLDO, to efficiently decompose and optimize large-scale complex topological overlapping problems. By leveraging the interactive learning capabilities of

reinforcement learning, RLDO learns the optimal decomposition policy from various decomposition results across various overlapping problems, in contrast to traditional methods that rely on predefined rules or random decomposition. To this end, we propose a Transformer-based policy network that effectively captures the essential features of overlapping problems. Furthermore, we construct the required state-action-reward tuples for the reinforcement learning process and adapt the PPO algorithm to update the policy network. The decomposition strategy network is then seamlessly integrated into the optimization process of large-scale overlapping problems with complex topologies.

Extensive experiments on both conflicting and conforming large-scale overlapping benchmark problems with complex topologies demonstrate that RLDO consistently outperforms state-of-the-art algorithms, including CBCCO, ORDG, RDG3, Random, and SaNSDE. In-depth analysis of the decomposition outcomes confirms RLDO's ability to analyze problem characteristics and generate effective decomposition strategies, leading to superior optimization performance.

In the future, we plan to extend RLDO into a dynamic decomposition-optimization framework that incorporates feedback from different optimization stages to further enhance the efficiency of the CC framework. An important future direction is to explore collaborative optimization strategies for handling subcomponents with conflicting objectives. Additionally, integrating the powerful reasoning capabilities of large-scale language models may offer a promising avenue to assist in solving increasingly complex large-scale overlapping problems.

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